12 November 2026

Dear Editor,

Please consider our manuscript, *Separating coverage gains from error targeting in molecular prediction sets*, as an original research article in *Chemometrics and Intelligent Laboratory Systems*. This is a substantially revised new submission following the decision on CHEMOLAB-D-26-01118.

The study introduces a framework for measuring the decision value of molecular prediction sets. Class-budget matching fixes the predictor and expansion count within each original singleton class, isolating error targeting from the amount and class allocation of deferral. An exact weighted-rank condition links transport weights to label addition and identifies when expansion reduces to probability ranking.

Across 960 predictor units spanning twelve Tox21 assays and random/scaffold holdouts from four additional assay families, the framework identifies a useful decision-aligned reference. On the historical six-assay panel, omitted-label probability ranking achieves 85.10% minimum-class coverage and 46.57% macro correct-singleton yield, compared with 84.75% and 44.61% for generic uncertainty at the same workload. It also reproduces the historical TAME coverage gain of 1.3649 percentage points. The activation analysis explains why transport-gate passage can coexist with zero intervention, connecting the mathematical rule to its observed molecular decisions.

The revised study combines these results with complete reference-system comparisons, paired molecular inference, and executable manuscript reconstruction. A point-by-point response details the changes from the earlier submission. The comparison framework and molecular evidence address chemometric prediction, uncertainty assessment, and the practical value of model complexity.

Siyuan Tong and Yuechen Wang contributed equally to this work and share first authorship. Siyuan Tong remains the corresponding author.

Thank you for considering our manuscript.

Sincerely,

Siyuan Tong

On behalf of the authors

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